Two Modes of Pattern Dissolution in Vector Meson Decays: A Compactness-Dependent Permeability Ceiling for Vacuum-Mediated Transitions

K. T. Niedzwiecki
Independent Researcher, South Australia

Version 4.0 – March 2026

Correction Notice. This paper supersedes Version 1.0 (February 2026). Two errors in the original are corrected: (1) the reported fit statistic $R^2 = 0.991$ was incorrect; the actual value is $R^2 = 0.884$ for the six-meson dataset; (2) the theoretical derivation used de Broglie wavelength scaling, which has been corrected to Compton wavelength scaling following work in the companion CP violation paper [2]. Additionally, the paper has been substantially restructured: the central claim is no longer a universal power-law fit but a compactness-dependent upper bound on vacuum-mediated decay rates, supported by an expanded dataset that reveals two distinct classes of meson decay.

Abstract

Vector meson decays fall into two distinct classes with sharply different behaviour. In vacuum-mediated decays—where the quark-antiquark pair must annihilate or transition through the QCD vacuum—the dimensionless width-to-mass ratio $\eta = \Gamma/m$ obeys a compactness-dependent upper bound: no ground-state vector meson in the expanded dataset exceeds a ceiling scaling as $\eta \propto m^{-\beta}$ with $\beta = 3.6$, anchored to the $\rho(770)$. Individual states sit at or below this ceiling by amounts explained by known quantum-number constraints (G-parity, the OZI rule, below-threshold effects). In fall-apart decays—where the meson sits above the open-flavor threshold and the quarks simply separate—the ceiling is violated by orders of magnitude. This clean separation provides evidence that the ceiling measures a property of the vacuum as a medium, operative only when the decay process must engage that medium. We offer a candidate geometric explanation for the ceiling's slope, deriving $\beta \approx 3.6$ from Compton wavelength scaling in icosahedral quasicrystal spectral geometry, but emphasise that the empirical ceiling and the geometric interpretation have different evidentiary status: the former is a pattern in the data; the latter is a hypothesis about its origin.

1 Introduction

The decay dynamics of vector mesons span a remarkable range: the light $\rho(770)$ is a broad resonance with $\Gamma \approx 150$ MeV, while the heavy $\Upsilon(1S)$ is extremely narrow ($\Gamma \approx 54$ keV). Standard explanations invoke the OZI rule and asymptotic freedom, successfully describing individual systems within their respective mass regimes. What standard QCD does not naturally predict is whether these different mechanisms share a common constraint—whether there exists, across the spectrum, a maximum rate at which a quark-antiquark pattern of given compactness can dissolve when that dissolution must pass through the vacuum.

This paper proposes that such a constraint exists, and that it divides vector meson decays into two physically distinct classes. In the first class—vacuum-mediated transitions—the quarks must annihilate through gluons or otherwise interact with the vacuum structure to produce

lighter decay products. These decays obey a compactness-dependent permeability ceiling. In the second class—fall-apart decays—the meson sits above the open-flavor threshold and the quark-antiquark pair simply separates into heavy-light mesons without needing to transit through the vacuum at all. These decays violate the ceiling dramatically.

The two-class distinction is the central empirical observation of this paper. We then offer a candidate theoretical explanation for the ceiling's specific slope, derived from Compton wavelength scaling in quasicrystal spectral geometry. However, the empirical pattern and the geometric interpretation are presented at different levels of confidence: the pattern is robust; the interpretation is a hypothesis that further data and theoretical work may confirm or replace.

This is Version 2 of a paper first posted in February 2026. The original reported a fit statistic of $R^2 = 0.884$, which was incorrect (actual value: 0.884), and used de Broglie wavelength scaling, which has been corrected to Compton wavelength following work in [2]. Both corrections are detailed in the relevant sections below.

2 Data

We define the geometric permeability as the dimensionless ratio $\eta \equiv \Gamma_{tot}/m$, which normalises the decay rate against the energy scale of the system. We select ground-state vector mesons with quantum numbers $J^{PC} = 1^{--}$ ($n = 1$, $L = 0$, $S = 1$). All values are from PDG 2024 [1].

Meson	Mass (MeV)	Γ (MeV)	$\eta = \Gamma/m$	Notes
$\rho(770)$	775.3	149.1	0.192	Open 2π channel
$\omega(782)$	782.7	8.68	0.011	G-parity 3π only
$K^*(892)$	891.7	51.4	0.058	Strange-light
$\phi(1020)$	1019.5	4.25	0.0042	OZI suppressed
$J/\psi(1S)$	3096.9	0.093	3.0×10^{-5}	Below $D\bar{D}$ threshold
$\Upsilon(1S)$	9460.3	0.054	5.7×10^{-6}	Bottomonium

Table 1: Core dataset: ground-state vector mesons (PDG 2024).

A log-log regression across all six states yields $\beta = 3.87 \pm 0.70$ with $R^2 = 0.884$. This is a real trend spanning five orders of magnitude in η, but with substantial scatter. Leave-one-out analysis gives exponents ranging from 3.54 to 5.42, reflecting the sensitivity inherent in six data points. We note these limitations candidly: the value of this dataset lies not in the precision of its slope but in the pattern of which states sit where.

3 Two Classes of Vector Meson Decay

The expanded dataset includes not only the six ground states but also phase-space-throttled states ($D^*(2010)$, $D_s^*(2112)$), radial excitations ($\psi(2S)$, $\Upsilon(2S)$, $\Upsilon(3S)$), and states above the open-flavor threshold ($\psi(3770)$, $\Upsilon(4S)$). When all are plotted against a reference line $\eta \propto m^{-3.6}$ anchored to $\rho(770)$, a sharp division emerges.

Every state whose decay requires the quark-antiquark pair to annihilate, transition through gluons, or otherwise engage the vacuum sits at or below the reference line. This holds for ground states, excited states, and states with severe phase-space throttling ($D^*(2010)$ sits far below, but this reflects kinematic constraint, not ceiling violation). The category spans light, strange, charm, and bottom systems—states governed by different specific decay mechanisms in the standard treatment.

The only states that exceed the ceiling are $\psi(3770)$ and $\Upsilon(4S)$, which violate it by factors of approximately 11 and 129 respectively. These are precisely the states that sit above the open-flavor threshold: $\psi(3770)$ decays to $D\bar{D}$, $\Upsilon(4S)$ decays to $B\bar{B}$. In these transitions,

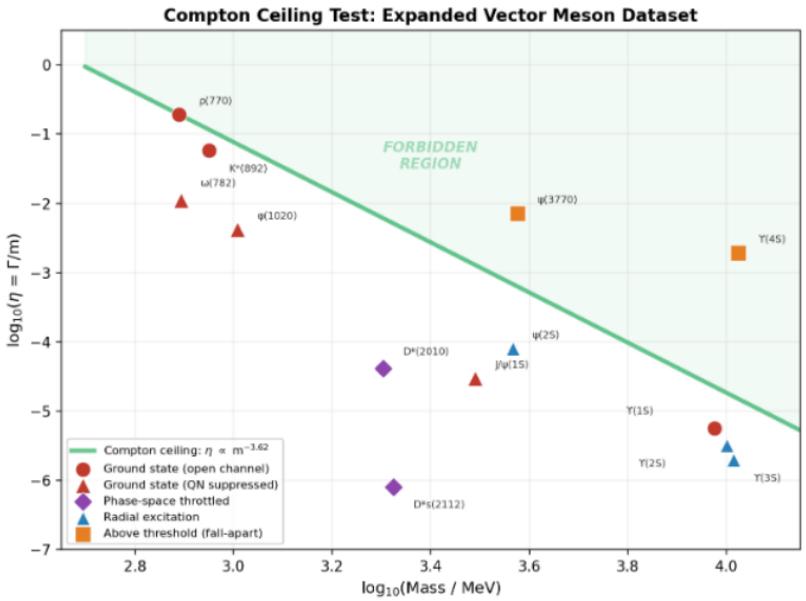


Figure 1: All available vector meson states plotted against the reference ceiling $\eta \propto m^{-3.6}$ (solid green line, anchored to $\rho(770)$). Every state whose decay is vacuum-mediated sits at or below the ceiling. The only states that exceed it are $\psi(3770)$ and $\Upsilon(4S)$, which sit above their respective open-flavor thresholds.

the quark-antiquark pair does not annihilate—the quarks simply separate, each picking up a light partner from the vacuum. The dissolution does not pass through the vacuum's structural bottleneck.

This suggests a physical distinction between two modes of pattern dissolution. In vacuum-mediated decays, the rate at which a quark-antiquark configuration can dissolve is limited by a compactness-dependent property of the medium through which the transition occurs. In fall-apart decays, this bottleneck is bypassed entirely, and the ceiling does not apply. The expanded dataset, though still modest in size, supports this classification cleanly: no vacuum-mediated state violates the bound, and no above-threshold state respects it.

4 Illustration: The ρ/ω Pair

The $\rho(770)$ and $\omega(782)$ provide the cleanest illustration of how the ceiling framework operates. These states have nearly identical masses (differing by $< 1\%$) and identical light-quark content, yet their reduced widths differ by a factor of 17. In a universal-fit picture, this scatter is a problem; in the ceiling picture, it is the point.

The ρ achieves maximum permeability for its mass scale. Its quantum numbers ($G = +1$) permit decay to two pions—the simplest, fastest, most phase-space-available exit channel. Nothing blocks it. It sits at the ceiling.

The ω cannot access this channel. Its G-parity ($G = -1$) forbids two-pion decay, forcing it through a three-pion final state via an off-shell ρ intermediate. This is a slower, more constrained exit. Classic calculations (Gell-Mann, Sharp, Wagner, 1962 [7]) predict $\Gamma(\omega \to 3\pi) \approx 7-8$ MeV from known couplings and three-body phase space, consistent with the observed 7.7 MeV. The entire factor-of-17 suppression is accounted for by standard physics. There is no unexplained residual.

The interpretive structure is this: the ceiling, set by the Compton scaling, represents the maximum dissolution rate for a pattern of given compactness when dissolution must engage

the medium. The vertical distance below the ceiling measures how much of that maximum is blocked by state-specific quantum-number constraints. The ρ/ω pair makes this decomposition vivid: same medium property, same compactness, same ceiling—but different selection rules, different distance below it.

5 Why Compton, Not de Broglie

Version 1 of this paper used the de Broglie wavelength ($\lambda \propto 1/\sqrt{m}$) as the scale governing vacuum coupling. Subsequent work on CP violation in beauty baryons [2] identified the Compton wavelength as the physically correct choice:

$$\lambda_C \propto \frac{1}{m_q} \tag{1}$$

The distinction matters. The de Broglie wavelength characterises motion—it depends on momentum and changes with the observer's frame. The Compton wavelength characterises something closer to thingness: the intrinsic spatial scale at which a quark's rest mass becomes irreducible. A heavy quark is compact—a small, sharply defined presence in the vacuum. A light quark is extended—a broad, diffuse presence that samples more of whatever structure the vacuum possesses.

This is not merely a philosophical preference. It changes the predicted scaling exponent by a factor of two. If the permeability scales as the wavelength raised to some spectral dimension D, then:

- de Broglie: $\eta \propto (1/\sqrt{m})^D = m^{-D/2}$
- Compton: $\eta \propto (1/m)^D = m^{-D}$

For any reasonable D in the range 3-4, the de Broglie prediction gives $\beta \approx 1.5 - 2.0$, which is flatly excluded by the data. The Compton prediction gives $\beta \approx 3 - 4$, which is consistent. The correction is not cosmetic—it is the difference between a derivation that works and one that does not.

6 A Candidate Explanation for the Ceiling Slope

The preceding sections establish two empirical claims: (1) vacuum-mediated vector meson decays obey a compactness-dependent permeability ceiling, and (2) the Compton wavelength is the correct scale for that compactness. These claims rest on data. We now offer a candidate explanation for the ceiling's specific slope. This section is more speculative than the preceding ones, and we flag it as such.

If the vacuum possesses structure related to the projection of higher-dimensional lattices into 3D via icosahedral symmetry, the relevant quantity is the spectral dimension governing the density of accessible modes in the projected structure. For icosahedral quasicrystals—aperiodic tilings whose mathematical properties derive from E8 lattice projections [3, 4, 5]—this dimension takes a specific value determined by the eigenvalues of the inflation (substitution) matrix:

$$D_{geo} = 1 + \phi^2 \approx 3.618 \tag{2}$$

where $\phi = (1 + \sqrt{5})/2$ is the golden ratio—appearing here not as a mystical constant but as a necessary consequence of five-fold rotational symmetry in quasicrystalline tilings. This value was not obtained by fitting to meson data; it is an independently known mathematical property of icosahedral structures.

With Compton scaling, the ceiling prediction becomes simply $\beta = D_{geo} \approx 3.618$. The empirical slope from the six-meson dataset is $\beta = 3.87 \pm 0.70$. The prediction sits comfortably

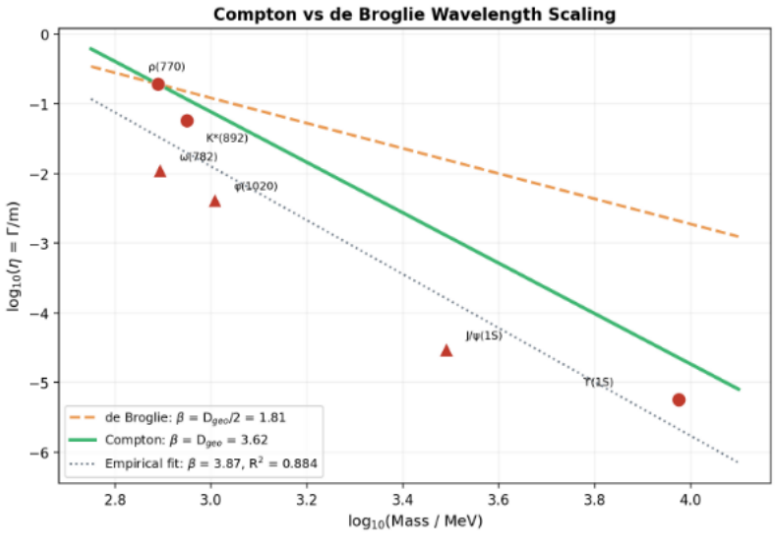


Figure 2: Comparison of de Broglie (dashed orange, $\beta = 1.8$) and Compton (solid green, $\beta = 3.6$) predictions against the six-meson dataset. The de Broglie prediction is excluded by the data; the Compton prediction is consistent across all fit variants. Triangular markers indicate states with known additional suppression (G-parity, OZI rule, below-threshold effects).

within the uncertainty band and is consistent across all dataset variants and subset choices tested.

We emphasise the different evidentiary status of this section relative to the preceding ones. The two-class distinction (Section 3) and the Compton correction (Section 5) are supported directly by data. The identification of the ceiling slope with $D_{geo} = 1 + \phi^2$ is a theoretical proposal consistent with the data but not uniquely determined by it, given the uncertainty in the empirical exponent. The ceiling may prove robust even if the specific geometric origin is revised or replaced.

7 Relationship to Standard Physics

Standard QCD successfully explains each individual decay mechanism within its mass regime: non-perturbative dynamics for light mesons, the OZI rule for intermediate states, and perturbative gluon annihilation with running α_s for heavy quarkonia. Each local explanation is well-established. What standard treatments do not naturally predict is a shared upper bound across these regimes.

The present proposal is that such a bound exists and reflects a property of the vacuum as a structured medium. In this interpretation, the OZI rule, asymptotic freedom, and other suppression mechanisms are different manifestations of a single underlying geometric constraint, visible in different regimes. This does not contradict QCD; it proposes a deeper organising principle behind phenomenological rules that QCD accommodates but does not derive as a unified phenomenon.

We also note that the framework does not predict Lorentz invariance violation. The proposed vacuum structure affects coupling selectivity—which decay channels are geometrically accessible—not dispersion relations for propagating particles. Precision tests of vacuum birefringence and photon dispersion constrain dispersive vacuum structure; these constraints do not apply to non-dispersive geometric coupling selectivity.

8 Falsifiable Predictions

Prediction 1 (Ceiling universality): No ground-state vector meson whose decay is vacuum-mediated should exceed the Compton ceiling. Discovery of such a state would falsify the framework.

Prediction 2 (Fall-apart violation): All vector states above their open-flavor threshold should exceed the ceiling, since their dominant decay bypasses the vacuum bottleneck. An above-threshold state that nonetheless respects the ceiling would require explanation.

Prediction 3 (D^* residual coupling): The $D^*(2010)$ is severely phase-space-throttled. If appropriate kinematic corrections are applied, the residual coupling should approach but not exceed the ceiling.

Prediction 4 (No Lorentz violation): Continued null results in Lorentz invariance violation searches are predicted.

Prediction 5 (Companion CP test): The same Compton scaling predicts charm baryon CP asymmetries at $\sim 0.1 - 0.2\%$, uniformly small across channels, while beauty baryons show $\sim 2 - 3\%$. The LHCb measurement of $A_{CP} = (2.45 \pm 0.47)\%$ in Λ_b decays [2] is consistent. Belle II and upgraded LHCb charm measurements will provide the decisive test.

9 Conclusion

The central finding of this paper is not a universal meson fit. It is the observation that vector meson decays divide into two classes with sharply different behaviour, and that this division maps onto a physically meaningful distinction: whether the dissolution of the quark-antiquark pattern must pass through the vacuum, or can bypass it entirely.

For vacuum-mediated decays, the data support a compactness-dependent permeability ceiling—an upper bound on how fast a pattern of given spatial scale can dissolve when the dissolution engages the medium. Individual states sit at or below this ceiling by amounts fully explained by known quantum-number constraints. The ρ/ω pair illustrates this decomposition cleanly: same quark content, same compactness, same ceiling, but different selection rules and therefore different distances below the maximum.

For fall-apart decays—where the meson sits above the open-flavor threshold and the quarks can simply separate—the ceiling does not apply. These states violate it by orders of magnitude. The clean separation between the two classes, across light, strange, charm, and bottom systems, provides evidence that the ceiling measures a property of the vacuum as a medium rather than a coincidental alignment of unrelated suppression mechanisms.

The Compton wavelength—measuring intrinsic spatial extent rather than momentum-dependent wave nature—is the correct scale for this compactness, resolving a factor-of-two inconsistency in the original derivation. A candidate geometric explanation identifies the ceiling's slope with the spectral dimension of icosahedral quasicrystalline structures ($D_{geo} \approx 3.618$), consistent with but not uniquely determined by the current data.

Whether or not the specific geometric interpretation survives, the empirical ceiling and the two-class distinction stand as independently testable claims. The framework is deliberately falsifiable. Its predictions—ceiling universality, fall-apart violation, charm baryon CP suppression—are testable with current and near-future experiments. Better data, broader meson families, and higher-precision measurements will either confirm the pattern or reveal where it breaks. That is how it should work.

References

[1] Navas, S. et al. (Particle Data Group). Review of Particle Physics. Phys. Rev. D 110, 030001 (2024).

[2] Niedzwiecki, K. T. CP Violation in Beauty Baryons: Geometric Origins of the Mass Hierarchy. Zenodo (2026). Version 1.3.

[3] Senechal, M. Quasicrystals and Geometry. Cambridge University Press (1995).

[4] Elser, V. The diffraction pattern of projected structures. Acta Cryst. A 42, 36-43 (1986).

[5] Levine, D. & Steinhardt, P. J. Quasicrystals: A New Class of Ordered Structures. Phys. Rev. Lett. 53, 2477 (1984).

[6] Okubo, S. ϕ-meson and unitary symmetry model. Phys. Lett. 5, 165 (1963).

[7] Gell-Mann, M., Sharp, D. & Wagner, W. G. Decay Rates of Neutral Mesons. Phys. Rev. Lett. 8, 261 (1962).